

REPOSYS - Campus Reprography Automation System

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Abstract

Reposys is a comprehensive web-based application developed to modernize and digitize the document printing and reprography services available in educational institutions. Traditional campus printing systems often rely on manual processes where students are required to stand in long queues, submit documents physically or through personal messaging applications such as whatsapp, wait for staff confirmation, and make cash payments. These practices lead to inefficient queue management, delays in service delivery, privacy concerns due to sharing personal documents on third-party platforms, payment inconveniences caused by the lack of exact change, and poor communication regarding order status.

Reposys addresses these challenges by providing a centralized digital platform that allows students, faculty, guests, counter staff, and administrators to efficiently manage the complete reprography workflow. Users can upload documents securely, configure printing options, estimate printing costs, make online or wallet-based payments, monitor live order status, and receive real-time notifications throughout the printing process. The system also supports guest users through a QR-code-based tracking mechanism without requiring account registration.

The proposed system follows a modern multi-tier architecture consisting of a React-based frontend, an Express.js backend, and MongoDB as the database. The application incorporates secure authentication using JSON Web Tokens (JWT), role-based access control, cloud-based document storage, Progressive Web App (PWA) capabilities for mobile accessibility, real-time communication using Socket.IO, and integrated online payment processing. Administrative modules provide centralized management of pricing, inventory, user accounts, service analytics, reports, and operational settings, while counter staff are provided with dedicated interfaces for order processing, queue management, and service updates.

A priority-based queue management mechanism is implemented to reduce waiting time while maintaining fairness among users. Additional modules including complaint management, digital wallet, push notifications, document analysis, QR-based order tracking, and automated inventory monitoring further enhance operational efficiency and user experience.

The primary objective of Reposys is to improve the efficiency, transparency, security, and accessibility of campus reprography services by replacing manual workflows with an integrated digital solution. The system minimizes document handling errors, protects user privacy, reduces unnecessary waiting time, enables cashless transactions, and provides complete visibility into every stage of the order lifecycle. By automating the entire reprography process, Reposys contributes toward building a smarter, more efficient, and digitally connected campus environment.

Technologies Used

Frontend

React.js | Vite | HTML5 | CSS3 | Tailwind CSS

Backend

Node.js | Express.js

Database

MongoDB Atlas

Mongoose ODM

Authentication & Security

JWT Authentication

Role-Based Access Control

Bcrypt Password Hashing

Cloud Services

Cloudinary

CloudConvert

Payment Integration

Razorpay

Real-Time Communication

Socket.IO